

AI-Assisted Software Development

Module 8b: Design and Architecture with AI Tools

Mike Burton 4-6 November 2025

Overview

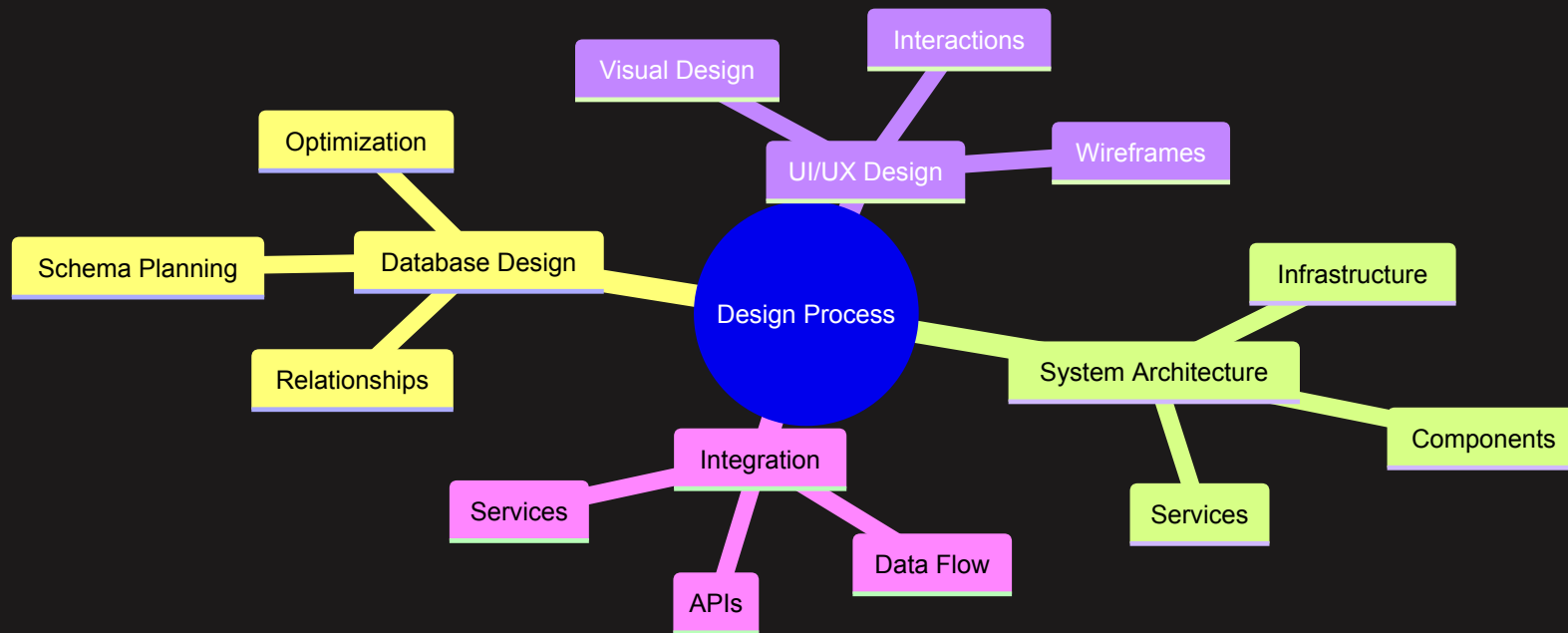
What we'll cover:

- Database Design
- System Architecture
- UI/UX Design
- Design Systems
- Technical Specifications

Tools we'll use:

- AI for architecture decisions
- V0 for UI generation
- Cursor for documentation
- AI for design validation

The Design Process



Database Design with AI

Start with your PRD and ask AI to:

1. Identify Data Entities

Core Entities:

- Users
- Projects
- Tasks
- Teams
- Comments

2. Define Relationships

Relationships:

- User belongs_to many Teams
- Team has_many Projects
- Project has_many Tasks
- Task has_many Comments

Database Schema Generation

Example AI conversation:

Input to AI:

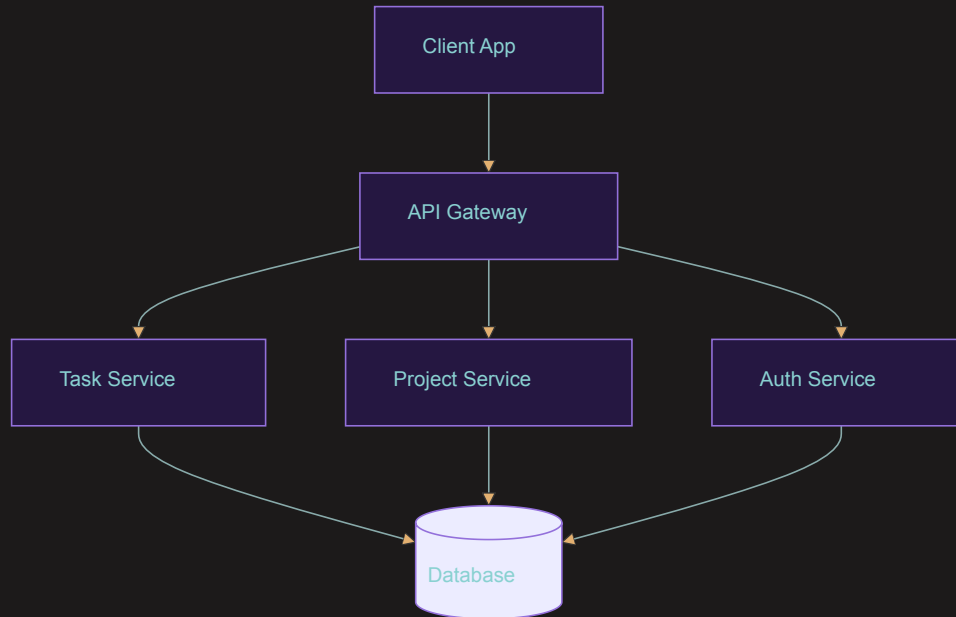
```
Generate a database schema for a
task management app with:
- User authentication
- Team collaboration
- Project organization
- Task dependencies
```

AI helps with:

- Table definitions
- Field types
- Relationships
- Indexes
- Constraints

System Architecture Planning

Use AI to design system components:



Break down into:

- Services
- Components
- Dependencies
- Data flow

UI/UX Design with V0

V0 workflow:

1. Start with Requirements

```
Design a task management app:  
- Clean, modern interface  
- Mobile-responsive  
- Team collaboration focus
```

2. Add Inspiration

- Reference similar apps
- Include screenshots
- Specify design preferences

3. Iterate Design

- Refine through prompts
- Focus on user experience
- Maintain consistency

UI Design Best Practices with V0

1. Clear Initial Prompts

Create a modern task dashboard with:

- Left sidebar navigation
- Card-based task view
- Status indicators
- Team member avatars

2. Iterative Refinement

Adjust the design to:

- Use more whitespace
- Increase contrast
- Add hover states
- Improve hierarchy

Design System Creation

Use AI to help define:

1. Visual Elements

- Colors
- Typography
- Spacing
- Icons

2. Components

- Buttons
- Forms
- Cards
- Navigation

3. Patterns

- Layouts
- Interactions
- Animations
- States

4. Documentation

- Usage guidelines
- Examples
- Best practices

Technical Specification Development

Convert designs to technical specs:

```
## Component: TaskCard
```

```
Properties:
```

- title: string
- description: string
- status: enum
- assignee: User
- dueDate: DateTime

```
Behaviors:
```

- Drag and drop
- Status toggle
- Quick edit
- Comment thread

Integration Planning

Plan how components work together:

1. API Design

```
POST /api/tasks
{
  "title": string,
  "project_id": uuid,
  "assignee_id": uuid,
  "due_date": datetime
}
```

2. State Management

```
{
  tasks: {
    byId: {},
    allIds: [],
    loading: boolean,
    error: string | null
  }
}
```

Design Validation

Use AI to validate designs for:

1. Usability

- Navigation flow
- User interactions
- Accessibility
- Mobile responsiveness

2. Technical Feasibility

- Performance impact
- Implementation complexity
- Resource requirements
- Scalability concerns

Documentation in Cursor

Maintain design documentation:

```
designs/  
├─ ui/  
│   ├── components/  
│   ├── layouts/  
│   └─ styles/  
├─ architecture/  
│   ├── services/  
│   └─ infrastructure/  
└─ database/  
    ├── schema/  
    └─ migrations/
```

Keep everything organized for AI context.

Best Practices

1. Design Process

- Start with user needs
- Design iteratively
- Validate early
- Document decisions

2. Tool Usage

- Provide clear context
- Use reference images
- Iterate designs
- Maintain consistency

3. Documentation

- Keep organized
- Update regularly
- Link related files
- Version control

Common Pitfalls

1. Design Phase

- ❌ Skipping user research
- ❌ Ignoring mobile views
- ✅ User-centered design
- ✅ Responsive first

2. Technical Planning

- ❌ Over-engineering
- ❌ Ignoring scalability
- ✅ Start simple
- ✅ Plan for growth

Recap

- Start with database design
- Plan system architecture
- Use V0 for UI design
- Create comprehensive design system
- Develop technical specifications
- Plan integration points
- Validate designs
- Maintain documentation

Next Steps

- Review designs with stakeholders
- Begin development setup
- Create component library
- Start implementation